

AI Usage Log: Sprint 1

SOEN 341: CareerNet

Feature: Login Backend

Branch: feature/login-backend

Pull Request: #34

Task 1

Prompt: Help with implementing the backend login functionality for the CareerNet team project by giving me a step by step instruction booklet on how to proceed.

Purpose of AI Use: Planning, implementation guidance and requirements analysis

How AI helped:

- Reviewed the existing FastAPI backend structure and registration implementation.
- Identified the files relevant to login.
- Explained the login flow: user lookup, password verification, JWT generation, and response.
- Kept the implementation consistent with the existing project structure.

Decision: Accepted

Reflection: This helped me understand how the login backend works as a whole before I started coding. I learned how the different parts connect together, especially finding the user in the database, verifying their password, generating a token, and returning the correct response. Having the process explained step by step also made it easier for me to work with the existing backend instead of creating something that did not match the rest of the project.

Task 2

Prompt: How to add the login request and response schemas to the existing authentication schemas?

Purpose of AI Use: Code generation and learning (request/response schemas)

How AI helped:

- Guided creation of LoginRequest with email and password.
- Guided creation of LoginResponse with access_token, token_type, and UserOut.
- Explained why UserOut prevents password information from being returned.
- Explained why login does not need to enforce the registration password-length rule.

Decision: Modified to suit our project

Reflection: I learned why the login endpoint needs separate request and response schemas and what information should be included in each one. I also understood why we return the public user information using UserOut instead of returning everything stored about the user. This helped me better understand how schemas control what data enters and leaves the backend.

Task 3

Prompt: Can you explain how to add JWT token generation to the existing password security code?

Purpose of AI Use: Code generation and learning (JWT authentication)

How AI helped:

- Extended the existing security.py while preserving Argon2 password hashing.
- Added environment-based JWT configuration.
- Added token expiration.
- Added the user's ID as the JWT "sub" claim.

Decision: Accepted

Reflection: Before this task, I did not fully understand what a JWT was or how it was used after a user logs in. I learned how the backend can generate a token with an expiration time and use the user's ID to identify them. I also learned that sensitive configuration should come from environment variables instead of being directly written in the code.

Task 4

Prompt: User: Can you help implementing the POST login endpoint in the existing authentication router?

Purpose of AI Use: Code generation and implementation guidance

How AI helped:

- Guided creation of POST /auth/login.
- Reused the existing database dependency and User model.
- Added case-insensitive email lookup and password verification.
- Used the same generic 401 error for unknown email and incorrect password.
- Returned the JWT and public user information.

Decision: Modified to match teammate's work

Reflection: This task helped me understand how all the previous pieces come together inside the actual login endpoint. I learned how the backend searches for the user, verifies the password, and creates the JWT before returning the response. I also learned why it is better to return the same error for an incorrect email and an incorrect password instead of revealing which one was wrong.

Task 5

Prompt: I need you to copy paste my prompts and then few bulletpoints of how you helped and numerate my prompts.

Purpose of AI Use: Documentation and formatting

How AI helped:

- Changed the log to an interaction-by-interaction format.
- Numbered the user prompts.
- Added concise bullet points explaining the AI assistance for each prompt.
- Distinguished exact prompts from earlier reconstructed prompts.

Decision: Modified to fit the format needed for the submission

Reflection: This helped me organize my AI usage in a clearer way and made me go back through the conversation to see exactly where AI helped me during the sprint. It also made me realize that I should document my AI use as I work instead of trying to remember everything at the end of the sprint.

Final Outcome

- Login backend implemented.
- JWT authentication added.
- Password verification integrated with existing Argon2 implementation.
- Five login tests added.
- All 17 backend tests passed.
- AI prompt documentation generated.